# A telescoping proof of Sun's harmonic-binomial series for $\zeta(3)$

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**Abstract**

We prove the identity in Conjecture 3.8, equation (3.12), of Zhi-Wei Sun's *Various conjectural series identities*. An explicit harmonic Wilf–Zeilberger pair reduces the inverse-binomial series to two elementary Euler sums. Its certificate is verified by two rational-function identities. A beta-integral estimate controls the entire remaining row of the telescoping array, including its harmonic factors. The proof uses only finite harmonic sums, exact polynomial algebra, positive double sums, and justified limits; numerical agreement is not used as a proof step. The accompanying scripts check the certificate exactly and independently enclose both sides through 120 decimal places.

## 1 Statement and scope

For an integer $m \geq 0$, put $H_0 = 0$ and $H_m = \sum_{j=1}^{m} j^{-1}$. Write $\zeta(3) = \sum_{j=1}^{\infty} j^{-3}$ and $\binom{2m}{m} = (2m)!/(m!)^2$.

**Theorem 1.** *The following series converges absolutely, and*

$$\boxed{\sum_{m=1}^{\infty} \frac{(21m-8)(2H_{3m-1}-H_{m-1}) - \dfrac{4(27m-11)}{3(3m-1)}}{m^3\binom{2m}{m}^3} = 3\zeta(3).} \tag{1}$$

This is exactly Sun's equation (3.12) in Conjecture 3.8 of [1], including the harmonic-number indices and the factor 3. Sun also posted this precise identity as formula (1) in [6]. Sun's Remark 3.8 relates it to Zeilberger's non-harmonic identity

$$\sum_{m=1}^{\infty} \frac{21m-8}{m^3\binom{2m}{m}^3} = \zeta(2).$$

The WZ acceleration method [2], and parameter expansions of WZ seeds [3, 4, 5], provide the methodological background. The successful deformation used to construct the present pair starts from Guillera's generalized Dougall summand [9]. Related derivative identities for the non-harmonic base kernel appear in Hou–Sun [10], Theorem 1.9; they have different harmonic factors from (1). Here these methods lead to a certificate that can be checked directly at integer indices. Consequently, the proof below requires no differentiation of an infinite series and no unevaluated hypergeometric transformation.

The latest revision displayed for Sun's source during the literature audit on 28 October 2026 was v3, dated 13 April 2026. A bounded search of later papers did not locate a proof of this exact identity. This is not a certification of priority or worldwide novelty. The identity is attributed to

Sun; this note supplies an explicit proof and reproducible checks. Other conjectures in that paper are outside its scope. The work was produced by GPT-6 Astra, an AI system, and internally checked; it is not represented as peer reviewed or formally verified in a proof assistant.

## 2 Two elementary Euler sums

The slow boundary of the telescoping array will involve two classical Euler sums; see [7] for the first. The triangular-sum proof of the second below adapts the elementary argument in [8]. We include all steps to avoid importing special-value evaluations.

**Lemma 1.**

$$\sum_{m=1}^{\infty} \frac{H_m}{m^2} = 2\zeta(3), \qquad \sum_{m=1}^{\infty} \frac{H_{2m}}{m^2} = \frac{11}{4}\zeta(3). \tag{2}$$

*Proof.* All double sums used here have nonnegative terms, so Tonelli's theorem permits their rearrangement, initially even before finiteness is known. Define

$$T = \sum_{a,b\geq 1} \frac{1}{ab(a+b)}.$$

Expanding $-\log(1-x) = \sum_{a\geq 1} x^a/a$ for $0 < x < 1$ gives

$$\begin{aligned} T &= \int_0^1 \frac{\log^2(1-x)}{x}\,dx = \int_0^1 \frac{\log^2 y}{1-y}\,dy \\ &= \sum_{r=0}^{\infty} \int_0^1 y^r \log^2 y\,dy = 2\sum_{r=0}^{\infty} \frac{1}{(r+1)^3} = 2\zeta(3). \end{aligned}$$

The last integral follows, for example, by substituting $y = e^{-u}$ and integrating $u^2 e^{-(r+1)u}$ twice by parts. For each $a \geq 1$, partial fractions and telescoping give

$$\sum_{b\geq 1} \frac{1}{b(a+b)} = \frac{H_a}{a}.$$

Thus $T = \sum_{a\geq 1} H_a/a^2$, proving the first formula.

Split $T$ into the diagonal $a = b$ and the two equal off-diagonal parts. With $a = n$ and $b = n + m$ in one off-diagonal part, this gives

$$U := \sum_{n,m\geq 1} \frac{1}{n(n+m)(2n+m)} = \frac{1}{2}\left(T - \frac{1}{2}\zeta(3)\right) = \frac{3}{4}\zeta(3).$$

The pointwise identity

$$\frac{2}{nm(2n+m)} = \frac{1}{nm(n+m)} + \frac{1}{n(n+m)(2n+m)}$$

therefore implies

$$V := \sum_{n,m\geq 1} \frac{1}{nm(2n+m)} = \frac{T+U}{2} = \frac{11}{8}\zeta(3).$$

On the other hand, $\sum_{m\geq 1} 1/[m(2n+m)] = H_{2n}/(2n)$, so $V = \frac{1}{2}\sum_{n\geq 1} H_{2n}/n^2$. This proves the second formula. The finite values obtained also establish convergence of every sum above. □

## 3 An explicit harmonic telescoping pair

All indices $n, k$ in this section are nonnegative integers. Define

$$f(n,k) = \frac{(n!)^4 (n+k)!^2}{(2n)!(2n+k+1)!^2} \tag{3}$$

and the finite harmonic combination

$$\begin{aligned} L(n,k) = & -\frac{5}{3}H_n + \frac{4}{3}H_{2n} + \frac{2}{3}H_{n+k} - \frac{4}{3}H_{2n+k+1} \\ & + H_{3n+2k+1} + H_{3n+2k+2}. \end{aligned} \tag{4}$$

The rational functions required for the certificate are

$$R(n,k) = \frac{P(n,k)}{2(2n+1)(k+2n+2)^2}, \tag{5}$$

$$S(n,k) = \frac{(k+n+1)Q(n,k)}{6(2n+1)^2(k+2n+2)^3(2k+3n+2)(2k+3n+3)}, \tag{6}$$

where the complete polynomials are

$$\begin{aligned} P(n,k) = & \, 2k^3 + (13n+11)k^2 + (28n^2+48n+20)k \\ & + 21n^3 + 55n^2 + 47n + 13, \end{aligned} \tag{7}$$

$$\begin{aligned} Q(n,k) = & \, 32k^5 + (328n+284)k^4 \\ & + (1328n^2+2316n+996)k^3 \\ & + (2646n^3+6961n^2+6036n+1721)k^2 \\ & + (2574n^4+9063n^3+11855n^2+6817n+1451)k \\ & + 972n^5+4290n^4+7512n^3+6516n^2+2796n+474. \end{aligned} \tag{8}$$

Every denominator in these expressions is positive on the stated index set. Put

$$\mathcal{F}(n,k) = f(n,k)L(n,k), \qquad \mathcal{G}(n,k) = f(n,k)\big(R(n,k)L(n,k) + S(n,k)\big). \tag{9}$$

**Lemma 2.** *The pair* (9) *satisfies*

$$\mathcal{F}(n+1,k) - \mathcal{F}(n,k) = \mathcal{G}(n,k+1) - \mathcal{G}(n,k). \tag{10}$$

*Proof.* This is an exact algebraic identity. For a transparent verification, write

$$A(n,k) = \frac{f(n+1,k)}{f(n,k)} = \frac{(n+1)^4(n+k+1)^2}{(2n+1)(2n+2)(2n+k+2)^2(2n+k+3)^2}, \tag{11}$$

$$B(n,k) = \frac{f(n,k+1)}{f(n,k)} = \frac{(n+k+1)^2}{(2n+k+2)^2}. \tag{12}$$

The harmonic increments $u = L(n+1,k) - L(n,k)$ and $v = L(n,k+1) - L(n,k)$ are rational:

$$\begin{aligned} u = & -\frac{5}{3(n+1)} + \frac{4}{3}\left(\frac{1}{2n+1} + \frac{1}{2n+2}\right) + \frac{2}{3(n+k+1)} \\ & - \frac{4}{3}\left(\frac{1}{2n+k+2} + \frac{1}{2n+k+3}\right) + \sum_{j=2}^{4} \frac{1}{3n+2k+j} + \sum_{j=3}^{5} \frac{1}{3n+2k+j}, \end{aligned} \tag{13}$$

$$v = \frac{2}{3(n+k+1)} - \frac{4}{3(2n+k+2)} + \sum_{j=2}^{3} \frac{1}{3n+2k+j} + \sum_{j=3}^{4} \frac{1}{3n+2k+j}. \tag{14}$$

Substitution of (5)–(8) gives the two rational identities

$$A - 1 = BR(n, k+1) - R(n, k), \tag{15}$$
$$Au = BR(n, k+1)v + BS(n, k+1) - S(n, k). \tag{16}$$

Both can be verified by multiplying through by their displayed denominators and collecting coefficients; each residual numerator is the zero polynomial in $\mathbb{Q}[n, k]$. All ingredients are printed above, so these checks require no external mathematical data. The supplied exact verifier performs these coefficient checks as well.

After division by $f(n, k) > 0$, the left side of (10) is $(A-1)L + Au$. The right side is $[BR(n, k+1) - R(n, k)]L + BR(n, k+1)v + BS(n, k+1) - S(n, k)$. Equations (15)–(16) prove equality. □

## 4 Convergence, boundaries, and evaluation

**Lemma 3.** *For every $n \geq 0$, $\sum_{k\geq 0} |\mathcal{F}(n, k)|$ converges and $\lim_{k\to\infty} \mathcal{G}(n, k) = 0$. Moreover,*

$$\lim_{n\to\infty} \sum_{k=0}^{\infty} |\mathcal{F}(n, k)| = 0.$$

*Proof.* The beta integral gives

$$f(n, k) = \frac{1}{\binom{2n}{n}} \left( \int_0^1 t^n (1-t)^{n+k}\, dt \right)^2.$$

Since $0 \leq t(1-t) \leq 1/4$, it follows that

$$0 < f(n, k) \leq \frac{16^{-n}}{\binom{2n}{n}(k+1)^2} \leq \frac{16^{-n}}{(k+1)^2}. \tag{17}$$

The sum of the absolute coefficients in (4) is 7. Using $H_j \leq 1 + \log(j+1)$ and $3n + 2k + 3 \leq (3n+3)(k+1)$ yields

$$|L(n, k)| \leq 7\big(1 + \log(3n+3) + \log(k+1)\big).$$

Consequently,

$$\sum_{k\geq 0} |\mathcal{F}(n, k)| \leq 7\, 16^{-n} \big[(1 + \log(3n+3))\zeta(2) + C\big], \qquad C = \sum_{j\geq 1} \frac{\log j}{j^2} < \infty. \tag{18}$$

The finiteness of $C$ follows from the integral test. This proves absolute row convergence and the asserted vanishing as $n \to \infty$.

For fixed $n$, the numerator and denominator degrees in $k$ in (5) and (6) show that $R(n, k) = O_n(k+1)$ and $S(n, k) = O_n(k+1)$. Equations (9) and (17) therefore give $\mathcal{G}(n, k) = O_n((1 + \log(k+1))/(k+1)) \to 0$. □

*Proof of Theorem 1.* Sum (10) over $0 \leq k \leq K$, and then let $K \to \infty$. Lemma 3 justifies the limit and gives

$$\mathcal{G}(n, 0) = \sum_{k\geq 0} \mathcal{F}(n, k) - \sum_{k\geq 0} \mathcal{F}(n+1, k).$$

Sum this equality over $0 \leq n < N$ and let $N \to \infty$. The terminal row vanishes by (18), so

$$\sum_{n=0}^{\infty} \mathcal{G}(n, 0) = \sum_{k=0}^{\infty} \mathcal{F}(0, k). \tag{19}$$

No exchange of two unbounded sums is needed in this argument.

We identify the fast boundary first. At $k = 0$, direct simplification gives

$$\begin{aligned} R(n,0) &= \frac{21n+13}{8(2n+1)}, \\ S(n,0) &= \frac{162n^2+229n+79}{24(2n+1)^2(3n+2)}, \\ L(n,0) &= 2H_{3n+2} - H_n - \frac{1}{3n+2} - \frac{4}{3(2n+1)}. \end{aligned}$$

Writing $m = n+1$, the factorial factor satisfies

$$f(n,0) = \frac{8(2n+1)}{m^3\binom{2m}{m}^3}.$$

The remaining rational simplification is

$$-\frac{1}{3n+2} - \frac{4}{3(2n+1)} + \frac{S(n,0)}{R(n,0)} = -\frac{4(27n+16)}{3(3n+2)(21n+13)}. \tag{20}$$

Thus $\mathcal{G}(n,0)$ is exactly the $m$th summand in (1).

For completeness, its absolute convergence can also be checked independently of the telescoping argument. If $B_m$ denotes the numerator in (1), then $0 < B_m < 126m^2$: indeed $2H_{3m-1} - H_{m-1} \geq H_{3m-1} \geq 3/2$, the subtracted rational quantity lies between 0 and 12, and $2H_{3m-1} - H_{m-1} \leq 2(3m-1)$. The central binomial coefficient is the largest of the $2m+1$ coefficients whose sum is $4^m$, hence $\binom{2m}{m} \geq 4^m/(2m+1)$. Therefore

$$0 < \frac{B_m}{m^3\binom{2m}{m}^3} < 3402\, m^2\, 64^{-m}. \tag{21}$$

For the slow boundary in (19), put $j = k+1$. Then $f(0,k) = j^{-2}$ and

$$L(0,k) = 2H_{2j} - \frac{2}{3}H_j - \frac{7}{6j}.$$

Lemma 1 now evaluates it as

$$\sum_{k=0}^{\infty} \mathcal{F}(0,k) = 2\left(\frac{11}{4}\zeta(3)\right) - \frac{2}{3}(2\zeta(3)) - \frac{7}{6}\zeta(3) = 3\zeta(3).$$

Together with the fast-boundary identification and (19), this proves (1). □

# 5 Exact checks and reproducibility

The mathematical proof is the certificate and boundary argument above. The computations in this section provide additional checks of its transcription and implementation.

## 5.1 Symbolic certificate

The verification scripts enter $P, Q, A, B, u, v$ as exact rational expressions and reduce the residuals of (15), (16), and (20) to zero. This proves identities of rational functions, not merely agreement on a finite grid. The denominators' positivity ensures their validity at every nonnegative integer pair $(n,k)$. An independent checker derives $u, v$ from the harmonic-index shifts and also checks concrete finite telescoping rectangles using rational arithmetic. The reported verification does not constitute a Lean or other proof-assistant formalization of the analytic argument.

### 5.2 Independent rational enclosures

The first term of (1) is 85/24. If $S_N$ is its sum through $m = N$, (21) bounds the omitted positive tail by

$$3402r^{N+1}\left(\frac{(N+1)^2}{1-r}+\frac{2(N+1)r}{(1-r)^2}+\frac{r(1+r)}{(1-r)^3}\right), \qquad r=\frac{1}{64}. \tag{22}$$

Every quantity here is rational.

For an independent enclosure of $3\zeta(3)$, the positive Euler transform gives

$$3\zeta(3)=4\sum_{q=0}^{\infty}\frac{d_q}{2^{q+1}}, \qquad d_q=\frac{1}{2}\int_0^1(1-t)^q\log^2 t\,dt=\frac{H_{q+1}^2+H_{q+1}^{(2)}}{2(q+1)}.$$

Here $H_j^{(2)}=\sum_{i=1}^{j} i^{-2}$ and $0<d_q\le 1$. To justify the formula without an alternating-sum interchange, sum the positive geometric series under the integral to obtain $\sum d_q/2^{q+1}=\frac{1}{2}\int_0^1 \log^2 t/(1+t)\,dt$. Expanding $1/(1+t)$ is justified by absolute integrability of the sum of the integrated absolute terms, and gives $\sum_{j\ge1}(-1)^{j-1}/j^3=\frac{3}{4}\zeta(3)$. The expression for $d_q$ follows by differentiating twice the finite beta formula $\int_0^1 t^{x-1}(1-t)^q dt=q!/[x(x+1)\cdots(x+q)]$ at $x=1$; domination by $t^{-1/2}\log^2 t$ justifies these derivatives. After $M$ terms, the positive omitted tail is at most $4\cdot 2^{-M}$.

Using $N=80$ and $M=500$, the two exact rational enclosures have widths less than $10^{-135}$ and $10^{-149}$, respectively. They have the same first 120 decimal digits. For orientation, the common value begins

$$3.606170709478782856199214484534349972294958870214966453768\ldots.$$

Such finite agreement alone would not prove an identity; it is separate from Theorem 1's proof.

### 5.3 Rebuilding the package

The accompanying package includes this source, both exact certificate checkers, the rational-enclosure script, a proof map, and a dated literature review. From the project directory, run

```
python src/verify_all.py
python src/build_paper.py
```

The symbolic checks require Python with SymPy; the enclosure check uses only the Python standard library. The paper build requires a working pdfLaTeX installation with the standard packages named in the source. The build script writes the final paper to `output/research_paper.pdf`. The proof itself does not rely on local paths, external data files, or the availability of these programs.

## References


[1] Z.-W. Sun, *Various conjectural series identities*, arXiv:2603.29973v3 (13 April 2026), Conjecture 3.8, equation (3.12). `https://arxiv.org/abs/2603.29973`.

[2] T. Amdeberhan and D. Zeilberger, *Hypergeometric series acceleration via the WZ method*, Electronic Journal of Combinatorics **4** (1997), R3. `https://arxiv.org/abs/math/9804121`.

[3] K. C. Au, *Wilf-Zeilberger seeds and non-trivial hypergeometric identities*, Journal of Symbolic Computation **130** (2025), 102421; arXiv:2312.14051v4. `https://arxiv.org/abs/2312.14051`.

[4] Q.-H. Hou and Z.-W. Sun, *Taylor coefficients and series involving harmonic numbers*, arXiv:2310.03699v3 (10 February 2026). https://arxiv.org/abs/2310.03699.

[5] K. C. Au, *Discovering hypergeometric series with harmonic numbers via Wilf-Zeilberger seeds*, arXiv:2602.08721v3 (4 September 2026). https://arxiv.org/abs/2602.08721.

[6] Z.-W. Sun, *Three new conjectural series involving harmonic numbers*, MathOverflow question 507600, 30 January 2026, formula (1). https://mathoverflow.net/questions/507600.

[7] J. M. Borwein and D. M. Bradley, *On two fundamental identities for Euler sums*, preprint (5 January 2005), Section 2. https://carmamaths.org/jon/Preprints/Papers/ArXiv/Zeta21/z21.pdf.

[8] Donald Splutterwit, answer to *Proving an alternating Euler sum*, Mathematics Stack Exchange, 4 September 2020 (edited 26 August 2021). The double-sum argument is adapted here with attribution. https://math.stackexchange.com/a/3813567.

[9] J. Guillera, *Dougall's ${}_5F_4$ sum and the WZ-algorithm*, Ramanujan Journal **46** (2018), 667–675; arXiv:1611.04385v5 (3 June 2019). https://arxiv.org/abs/1611.04385.

[10] Q.-H. Hou and Z.-W. Sun, *Evaluations of some series via the WZ method*, arXiv:2604.15172v1 (16 April 2026), Theorem 1.9. https://arxiv.org/abs/2604.15172.